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# Creating Novel Virtual Reality Interfaces for Teaching Computer Architecture

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of the requirements for the degree of  
MSc in Computer Science: Augmented and Virtual Reality

# Declaration

I have read and I understand the plagiarism provisions in the General Regulations of the University Calendar for the current year, found at <http://www.tcd.ie/calendar>.

Signed: Priyansh Nayak

Date: December 20, 2025

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# 1 Research Topic Summary

## 1.1 Motivation for Research Topic

Computer architecture is widely recognized as one of the more challenging areas of computer science education due to the abstract and dynamic nature of its core concepts. Topics such as instruction execution, pipelining, memory hierarchy, and stack management require students to reason about processes that are largely invisible and difficult to conceptualize through static diagrams or text-based tools alone. This challenge was observed directly through my experience as an Undergraduate Teaching Assistant for a Computer Organization course over four semesters, where recurring student difficulties centered on forming accurate mental models of how architectural components interact during program execution.

In practice, student understanding often improved only after extensive visual explanation, such as walking through code execution on a whiteboard, tracing stack updates across function calls, or manually illustrating instruction flow through pipeline stages. While tools such as simulators and assembly environments provide functional correctness, they offer limited support for intuitive spatial and temporal understanding. Similar abstraction challenges have also been noted in other computer science domains, such as database systems, where learners struggle to visualize query execution and data flow.

Virtual Reality (VR) offers the potential to address these challenges by enabling immersive, interactive visualization of otherwise hidden processes. By allowing learners to explore and observe architectural behavior as it unfolds actively, VR may support deeper conceptual understanding than traditional teaching approaches.

## 1.2 Research Question

To what extent does an immersive virtual reality-based learning environment improve students' understanding of abstract computer architecture concepts compared to traditional teaching methods?

### 1.2.1 List of Research Objectives

1. To design and implement an immersive VR-based learning environment that visualizes selected abstract computer architecture concepts, such as instruction flow, pipelining, and stack behavior during program execution.

2. To develop learning scenarios within the VR environment that allow students to actively explore and interact with these architectural concepts as they evolve during code execution.
3. To evaluate the impact of the VR-based learning environment on students' conceptual understanding of computer architecture by comparing learning outcomes with those achieved through traditional teaching materials.
4. To assess student engagement and perceived usability of the VR learning environment using established evaluation instruments.
5. To identify design strengths, limitations, and potential improvements of using immersive VR for teaching abstract computer science concepts, informed by both quantitative results and qualitative student feedback.

### **1.2.2 Approach/Methods Planned to Achieve Research Objectives**

1. Review existing literature on the use of virtual reality for computer science education, with a focus on computer architecture learning and related abstract domains, to inform system design and evaluation choices.
2. Design and implement an immersive VR-based learning environment to visualise selected computer architecture concepts, such as instruction flow, pipelining, and stack behavior during program execution.
3. Develop guided learning scenarios within the VR environment that align with existing computer architecture teaching materials.
4. Conduct an evaluation study with student participants, using conceptual understanding assessments, engagement and usability questionnaires, and qualitative feedback to compare the VR-based approach with traditional teaching methods.

### **1.2.3 Summary of Evaluation**

The evaluation of this research will aim to examine the effectiveness of an immersive VR-based learning environment in supporting students' understanding of abstract computer architecture concepts. An appropriate evaluation strategy will be selected to allow comparison between the VR-based approach and existing teaching methods commonly used in computer architecture education.

The evaluation will consider both learning-related and experiential outcomes. Learning effectiveness will be assessed using measures designed to capture students' conceptual understanding of the selected computer architecture topics before and after interaction with

the learning materials. In addition, students' engagement with and perceived usability of the VR environment will be examined to provide complementary insight into their learning experience.

A combination of quantitative and qualitative data collection methods will be employed to provide a balanced evaluation of the proposed approach. Quantitative measures will be used to identify overall trends in learning outcomes, while qualitative feedback will support interpretation of the results and help identify strengths, limitations, and potential areas for improvement. This flexible evaluation approach is intended to support meaningful assessment of the VR-based learning environment while allowing refinement of the evaluation design as the research progresses.

#### **1.2.4 Anticipated Contributions**

This research is expected to contribute to the understanding of how immersive virtual reality can be used to support learning of abstract computer architecture concepts in computer science education. It will provide design and evaluation insights into the use of VR for visualizing dynamic system behavior that is difficult to convey using traditional teaching methods. The findings may inform the development of future VR-based educational tools and may also apply to other abstract computer science domains, such as database systems, where similar visualization challenges exist.

## 2 Literature Review Plan Summary

Concepts for literature review:

- (a) Virtual reality in education
- (b) Virtual reality in computer science education
- (c) Computer architecture education
- (d) VR-based learning for computer architecture
- (e) visualization of abstract computer science concepts
- (f) VR-based learning environments
- (g) Use of immersive technologies for teaching system-level concepts
- (h) Challenges and limitations of using VR in educational contexts
- (i) visualization and interaction design in VR
- (j) Transferability of VR-based learning approaches to related computer science domains

### 2.1 Keyword Search Summary

#### 2.1.1 Keyword Search Results

Seven references resulting from the keyword search, and their anticipated relevance:

- (i) [1]: relevant because it presents an immersive VR system designed specifically to support experiential learning of computer architecture concepts.
- (ii) [2]: relevant because it demonstrates a VR serious game that visualises internal CPU operations, closely aligning with the proposed research focus.
- (iii) [3]: relevant because it provides evidence that interactive and game-based approaches can improve learning outcomes in computer architecture education.
- (iv) [4]: relevant because it discusses the role of virtual reality in supporting the learning of abstract computer science concepts through interactive visualization.
- (v) [5]: relevant because it offers a systematic review of virtual reality applications in computer science education, providing broader contextual grounding.

- (vi) [6]: relevant because it identifies pedagogical and practical challenges associated with the adoption of VR in educational settings.
- (vii) [7]: relevant because it illustrates how VR can be used to visualise abstract system structures in database education, offering transferable design insights.

### 2.1.2 Keyword Search Approach

The keyword-based literature search was conducted using Google Scholar to identify academic work relevant to the use of virtual reality in computer science education, with a particular focus on computer architecture learning. An initial set of keywords included combinations of terms such as virtual reality, computer architecture education, computer science education, and immersive learning. These keywords were refined to include more specific phrases such as VR-based learning, serious games, and visualization of abstract concepts to reduce the number of unrelated technical results.

The search was limited to peer-reviewed journal articles, conference papers, and systematic reviews published in recognized academic venues. Sources were evaluated based on relevance to educational outcomes, credibility of the publication venue, and alignment with the research topic. Papers that focused primarily on hardware performance, low-level system implementation, or non-educational VR applications were excluded. This filtering process resulted in a focused set of references that directly address immersive and interactive approaches to teaching computer architecture and related computer science concepts.

## 2.2 AI Search Summary

### 2.2.1 AI Search Results

Seven references resulting from the AI-assisted search, and their anticipated relevance:

- (i) [4]: relevant because it is a broadly applicable discussion of virtual reality in computer science education, commonly surfaced by AI-based searches.
- (ii) [5]: relevant because systematic reviews are frequently prioritized by AI tools when identifying foundational literature.
- (iii) [8]: relevant because it provides a high-level overview of trends and challenges in VR-based education, despite limited focus on computer science.
- (iv) [9]: relevant because it discusses visualization challenges and design considerations in virtual reality environments.
- (v) [10]: relevant because it examines learning outcomes and technology acceptance in



VR, though its focus on pedagogical agents limits domain specificity.

- (vi) [11]: relevant because it presents VR-based simulation use cases that were returned by the AI search despite limited educational evaluation.
- (vii) [12]: relevant because it exemplifies an AI-suggested result that is technically focused and largely unrelated to educational objectives.

## 2.2.2 AI Search Approach

The AI-assisted literature search was conducted using ChatGPT to identify relevant research related to virtual reality in education and computer science learning. The initial prompt described the research focus on immersive VR-based learning environments, computer architecture education, and the visualization of abstract system behavior. The results were broad, prompting a refinement of the prompt to emphasize educational contexts and learning outcomes.

All AI-suggested references were manually verified to assess their relevance, credibility, and authoritativeness. Verification involved reviewing publication venues, authorship, and the stated research focus of each paper. Preference was given to peer-reviewed publications and recognized surveys, while papers that were primarily technical or infrastructure-focused were excluded. This process highlighted both the strengths and limitations of AI-assisted search, demonstrating its usefulness for discovering broad or adjacent literature while requiring careful human judgment to filter irrelevant results.

## 2.3 Comparison/Reflection

The keyword-based and AI-assisted search approaches produced overlapping but distinct sets of literature. The keyword search resulted in a more targeted collection of studies directly related to virtual reality and computer architecture education, reflecting the precision of carefully selected search terms. In contrast, the AI-assisted search returned a broader range of sources, including high-level educational reviews as well as technically oriented papers with limited relevance to learning objectives.

These differences are likely due to the structured constraints of keyword searches compared to the semantic and exploratory nature of AI-assisted tools. As a result, greater confidence would be placed in the keyword-based approach when identifying core literature for the dissertation. However, the AI-assisted approach can complement traditional searches by surfacing broader perspectives and adjacent work that may not be captured through keywords alone. Together, these methods support a more comprehensive and balanced literature review process.

## 3 Supporting Information

### 3.1 Research Ethics Considerations

This dissertation research is likely to require a School Ethics Application, as it involves the participation of human subjects in an educational evaluation study. The proposed research involves collecting data from students interacting with a virtual reality-based learning environment, which may include measures of conceptual understanding, engagement, usability feedback, and qualitative responses. As such, participants' consent, anonymity, and data protection must be carefully managed.

An ethics application would ensure that participants are fully informed about the purpose of the study, that participation is voluntary, and that they may withdraw at any time without penalty or consequence. All collected data would be anonymous and stored securely in accordance with institutional data protection policies. No vulnerable populations are anticipated to be involved, and the study poses minimal risk beyond those encountered in standard educational research.

### 3.2 Technology Ethics Considerations

If the proposed VR-based learning approach were widely adopted, several ethical considerations would need to be addressed. One concern relates to accessibility and inclusion, as not all students may be able to use immersive VR systems due to physical discomfort, motion sickness, or lack of access to appropriate hardware. Over-reliance on VR-based tools could disadvantage such students if alternative learning pathways are not provided.

There are also considerations around data privacy, as VR learning environments may collect interaction data to support evaluation or adaptation. Care must be taken to ensure that any collected data is limited to what is necessary, anonymous where possible, and used transparently. Additionally, immersive environments may increase cognitive load if poorly designed, potentially hindering learning rather than supporting it. Addressing these issues through careful design, optional use, and clear data governance would be essential to ensure that the technology benefits learners without introducing unintended harms.

### 3.3 Skills you aim to Acquire through Dissertation Research

Through this dissertation research, I aim to develop a combination of technical and research-oriented skills relevant to virtual reality development and computer science education. On the technical side, this project will allow me to gain practical experience in designing and implementing immersive VR environments, including interaction design, spatial visualization, and the representation of dynamic system-level processes such as instruction execution and pipelining. I also aim to strengthen my skills in using VR development tools and frameworks, as well as applying user-centered design principles within an educational context.

From a research perspective, the dissertation will provide experience in conducting educational research, including experimental design, evaluation planning, and data collection involving human participants. I aim to develop skills in analyzing both quantitative learning outcome data and qualitative user feedback, as well as critically interpreting evaluation results. In addition, this project will support the development of academic skills such as literature review, research writing, and reflective analysis. Together, these skills will contribute to my ability to design, evaluate, and communicate research involving immersive technologies in computer science education.

### 3.4 Pivots in Research Plan

Following discussions with my supervisor during this semester, an early pivot involved conducting a preliminary literature review earlier than originally planned, rather than deferring this work to a later stage. This enabled earlier refinement of the research area and supported the formulation of potential research questions. Based on my supervisor's advice, the research question was subsequently formalized.

A second pivot concerned the technical scope of the project. My supervisor recommended initially prototyping a lower-complexity system, such as a database-focused implementation, before potentially extending the work toward a more complex computer architecture-based system. This feedback informed the phased structure of the plan, allowing for early prototyping and evaluation prior to committing to a more technically demanding direction.

Additionally, feedback from a peer influenced the design approach of the proposed learning experience. While the original plan focused primarily on visualization for teaching computer architecture concepts, peer feedback suggested incorporating more interactive and potentially gamified elements, such as puzzle-based interactions. This refinement aimed to

enhance learner engagement and conceptual understanding and informed revisions to the conceptual design and prototyping stages of the plan.

### 3.5 Workplan

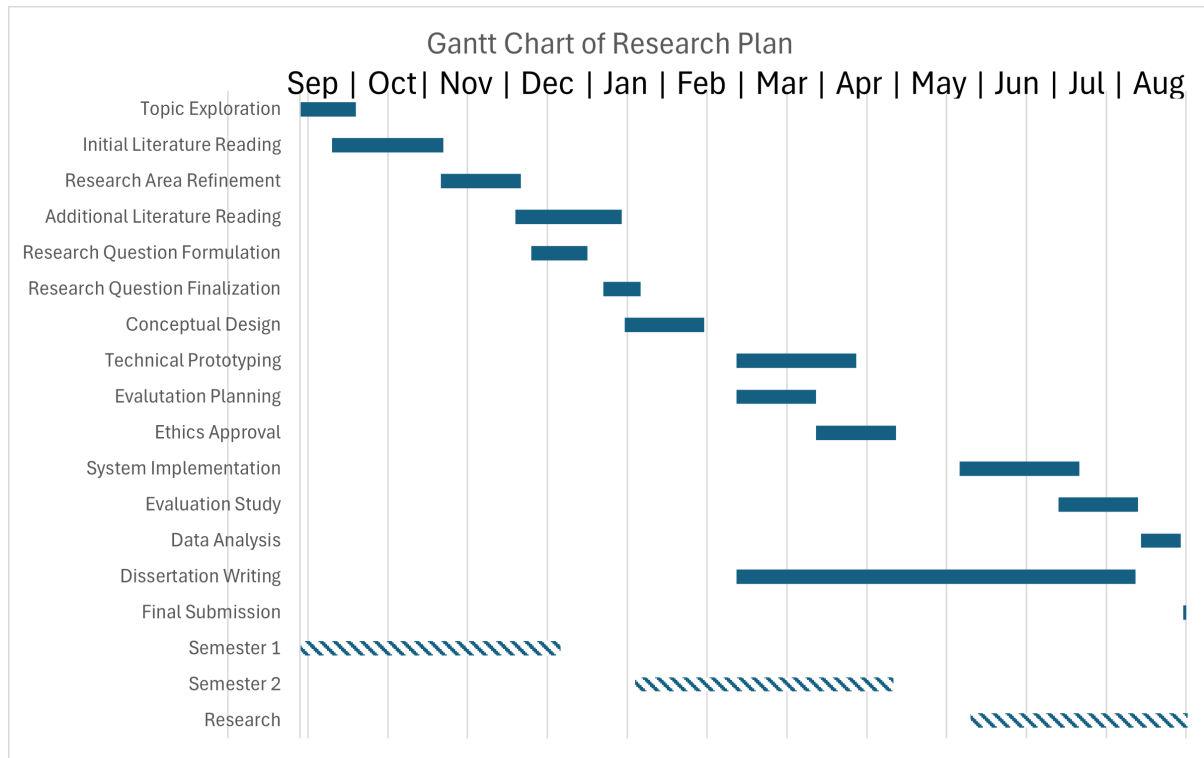


Figure 3.1: Gantt chart of dissertation project from start of semester 1 to submission date

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